

Rui Sousa

Resume

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WORK EXPERIENCE

JUN 2018 - SEP 2021 (RESEARCH ASSISTANT)

London School of Economics
Profs. Benjamin Moll & Ricardo Reis

- Solved continuous-time heterogeneous agent models using finite-difference methods.
- Analysed consumer and professional inflation expectations microdata
- Modeled rare inflation events using extreme value theory (threshold exceedances + Pareto tail estimation)

JUN 2018 - SEP 2021 (ANALYST)

European Central Bank
Crisis Management - Quantitative Cluster

- Contributed to early warning model using global media data (GDELT) and textual analysis (frequency, sentiment)
- Analysed liquidity provisions during extreme events
- Contributed to the impact assessment of Basel III on MREL requirements

JUN 2018 - JUN 2019 (TRAINEE)

European Central Bank
Stress Test Modelling

- Assisted with research assessing the impact of stress-tests and capital requirements on capital build-up and profitability using causal-inference methods (regression discontinuity, triple differences-in-differences)
- Contributed to credit risk assessment tool using historical loan portfolio data

SKILLS

TIME-SERIES ANALYSIS Structural Linear Models, Factor Models, Time-varying Volatility, Bayesian Inference, Forecasting

MICROECONOMETRICS Panel Data Analysis, Shift-share Instrumental Variables, Differences-in-Differences, Regression Discontinuity Designs, Dimension Reduction

MODELLING DSGE Models, Demand Estimation

PROGRAMMING: R, Python

COMPUTING: High-Performance Computing (SLURM, job arrays), Git, L^AT_EX

DOCTORAL RESEARCH

Working Papers

• The Aggregate Green Elasticity of Substitution (Draft)

- Estimated aggregate energy demand using shift-share instrumental variables. Found complementarity (elasticity < 1) between clean and fossil energy across US states.

Work In Progress

• Climate Change and the Sovereign Debt Overhang

- Estimate GDP impact of extreme climate events using local projections. Use results to calibrate sovereign default model under projected carbon pathways.

• The Distributional Effects of Climate Change

- Use income/wealth distributional data and identified temperature innovations to estimate the distributional impacts of climate change using local projections.

EDUCATION

2021 – 2027 (EXPECTED) **Northwestern University**
EVANSTON, IL
Doctor of Philosophy
Department of Economics

2016 – 2019 **Nova SBE**
LISBON, PORTUGAL
MSc in Economics
GPA: 19/20 (A)

2013-2016 **University of Porto**
PORTO, PORTUGAL
BSc in Economics
GPA: 18/20 (A)

AWARDS

2024-2025 **PhD Scholarship**
Fundação para a Ciência e a Tecnologia

2016-2019 **MSc Scholarship**
Nova SBE

2015 **Top 3 Students EU Commission Trip**
University of Porto

